Self-Assessment Multiple Choice Questions (MCQs) – Electromagnetism

Fall 2018

Name: ______________________ Roll No.:______________________ Section:________

1) In the formula $\vec{F} = q\vec{v} \times \vec{B}$:
A) $\vec{F}$ must be perpendicular to $\vec{v}$ but not necessarily to $\vec{B}$
B) $\vec{F}$ must be perpendicular to $\vec{B}$ but not necessarily to $\vec{v}$
C) $\vec{v}$ must be perpendicular to $\vec{B}$ but not necessarily to $\vec{F}$
D) all three vectors must be mutually perpendicular
E) $\vec{F}$ must be perpendicular to both $\vec{v}$ and $\vec{B}$.

2) A static magnetic field CANNOT:
A) exert a force on a charge
B) accelerate a charge
C) change the momentum of a charge
D) change the kinetic energy of a charge
E) exist

3) A conducting strip of width 1.5 mm is in a magnetic field. As a result, there is a potential difference of 4.3 mV across the width of the strip. What is the magnitude of the electric field in the strip?
A) 0.35 V/m
B) 1.2 V/m
C) 1.9 V/m
D) 2.9 V/m
E) 6.4 V/m

4) The Hall effect can be used to calculate the charge-carrier number density in a conductor. If a conductor carrying a current of 2.0 A is 0.5 mm thick, and the Hall effect voltage is 4.5 μV when it is in a uniform magnetic field of 1.2 T, what is the density of charge carriers in the conductor?
A) $1.0 \times 10^{28}/m^3$
B) $6.7 \times 10^{27}/m^3$
C) $4.6 \times 10^{27}/m^3$
D) $1.7 \times 10^{27}/m^3$
E) $1.2 \times 10^{27}/m^3$

5) The magnetic field outside a long straight current-carrying wire depends on the distance R from the wire axis according to:
A) R
B) $1/R$
C) $1/R^2$
D) $1/R^3$
E) $1/R^{3/2}$

6) Two long straight wires are parallel and carry current in opposite directions. The currents are 8.0 A and 12 A and the wires are separated by 0.40 cm. The magnetic field at a point midway between the wires is:
A) 0 T
B) 4.0×10^{-4} T
C) 8.0×10^{-4} T
D) 12×10^{-4} T
E) 20×10^{-4} T

7) In an overhead straight wire, the current is north. The magnetic field due to this current, at our point of observation, is:
A) east
B) up
C) north
D) down
E) west

8) Lines of the magnetic field produced by a long straight wire carrying a current are:
A) in the direction of the current
B) opposite to the direction of the current
C) leave the wire radially
D) circles concentric with the wire
E) lines similar to those produced by a bar magnet

9) A coulomb is:
A) one ampere per second
B) the quantity of charge which will exert a force of 1 N on a similar charge at a distance of 1 m
C) the amount of current in each of two long parallel wires separated by 1 m, which produces a force of 2×10^{-7} N per meter
D) the amount of charge which flows past a point in one second when the current is 1 A
E) an abbreviation for a certain combination of kilogram, meter and second

10) In Ampere's law, $\oint \vec{B} \cdot d\vec{s} = \mu_0 i$, the integration must be over any:

A) surface
B) closed surface
C) path
D) closed path
E) closed path that surrounds all the current producing $\vec{B}$

11) The magnetic field $\vec{B}$ inside a long ideal solenoid is independent of:

A) the current
B) the number of turns of wire
C) the spacing of the windings
D) the cross-sectional area
E) the direction of the current

12) A solenoid is 3.0 cm long and has a radius of 0.50 cm. It is wrapped with 500 turns of wire carrying a current of 2.0 A. The magnetic field at the center of the solenoid is:

A) 9.9×10^{-8} T
B) 1.3×10^{-3} T
C) 4.2×10^{-2} T
D) 16 T
E) 20 T

13) A toroid with a square cross section carries current i. The magnetic field has its largest magnitude:

A) at the center of the hole
B) just inside the toroid at its inner surface
C) just inside the toroid at its outer surface
D) at any point inside (the field is uniform)
E) at none of the above

14) The emf that appears in Faraday's law is:

A) around a conducting circuit
B) around the boundary of the surface used to compute the magnetic flux
C) throughout the surface used to compute the magnetic flux
D) perpendicular to the surface used to compute the magnetic flux
E) none of the above

15) The normal to a certain 1.0 m² area makes an angle of 60° with a uniform magnetic field. The magnetic flux through this area is the same as the flux through a second area that is perpendicular to the field if the second area is:

A) 0.50 m²
B) 0.87 m²
C) 1.0 m²
D) 1.2 m²
E) 2.0 m²

16) Suppose this page is perpendicular to a uniform magnetic field and the magnetic flux through it is 5.0 Wb. If the page is turned by 30° around an edge the flux through it will be:

A) 2.5 Wb
B) 4.3 Wb
C) 5.0 Wb
D) 5.8 Wb
E) 10 Wb

17) A 2.0 T uniform magnetic field makes an angle of 30° with the z axis. The magnetic flux through a 3.0 m² portion of the xy plane is:

A) 2.0 Wb
B) 3.0 Wb
C) 5.2 Wb
D) 6.0 Wb
E) 12 Wb

18) A rod of length L and electrical resistance R moves through a constant uniform magnetic field $\vec{B}$; both the magnetic field and the direction of motion are parallel to the rod. The force that must be applied by a person to keep the rod moving with constant velocity $\vec{v}$ is:

A) 0
B) BLv
C) BLv/R
D) B^2L^2v/R
E) $B^2L^2v^2/R$

19) 1 weber is the same as:

A) 1 V·s
B) 1 T·s
C) 1 T/m
D) 1 V/s
E) 1 T/m²

20) You push a permanent magnet with its north pole away from you toward a loop of conducting wire in front of you. Before the north pole enters the loop the current in the loop is:

A) zero
B) clockwise
C) counterclockwise
D) to your left
E) to your right